Beginning hypothesis testing

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Course status so far

So far:

- How can we estimate parameters / fit a model?
  - MLE, method of moments, M-estimation, and applications to regression
- What makes a good estimator?
  - MSE, unbiased estimators, CRLB, consistency, asymptotic relative efficiency
- What is the large-sample behavior of our estimators?
  - WLLN, CLT, delta method, asymptotic normality of the MLE, asymptotic normality of M-estimators

Next: how can we use our estimates for inference?

- Hypothesis tests using Normal distributions
- General hypothesis testing framework
- Other methods for testing hypotheses (Neyman-Pearson, LRT)
- Confidence intervals

Hypothesis tests for a population mean

Let $X_1, X_2, ..., X_n$ be an iid sample from a population with unknown mean μ, and known variance σ^2. We want to test

value hypothesized under H_0

$$H_0 : \mu = \mu_0 \qquad H_A : \mu \neq \mu_0$$

Test statistic:

$$Z_n = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}}$$

CLT: $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$

if H_0 is true, $Z_n = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}} \approx N(0,1)$

(null distribution)

Hypothesis tests for a population mean

Let $X_1, X_2, ..., X_n$ be an iid sample from a population with mean μ and variance σ^2. We want to test

$$H_0 : \mu = \mu_0 \qquad H_A : \mu \neq \mu_0$$

Test statistic: $Z_n = \dfrac{\overline{X}_n - \mu_0}{\sigma/\sqrt{n}}$

What if σ is unknown?

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2} \qquad s \xrightarrow{P} \sigma$$

$$T_n = \frac{\bar{X}_n - \mu_0}{s/\sqrt{n}} \xrightarrow{d} N(0,1)$$

Alternative: $T_n \approx t_{n-1}$

(as $\nu \to \infty$, $t_\nu \to N(0,1)$)

Hypothesis tests for a population mean

Let $X_1, X_2, ..., X_n$ be an iid sample from a population with mean μ and variance σ^2. We want to test

$$H_0 : \mu = \mu_0 \qquad H_A : \mu \neq \mu_0$$

Test statistic: $Z_n = \dfrac{\overline{X}_n - \mu_0}{\sigma/\sqrt{n}}$

Rejecting: Should we reject H_0 when Z_n is close to 0, or when Z_n is far away from 0?

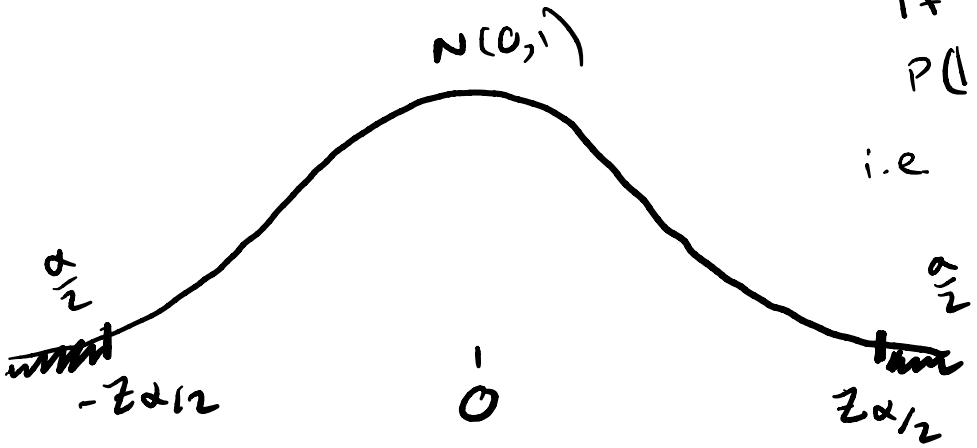
reject H_0 when $|Z_n| >$ cutoff

Approximate distribution of Z_n under H_0:

If $Z \sim N(0,1)$:

$P(|Z| > z_{\frac{\alpha}{2}}) = \alpha$

i.e $P(\text{reject } H_0 \mid H_0 \text{ true}) \approx \alpha$



Hypothesis tests for a population mean

Let $X_1, X_2, \ldots$ be an iid sample from a population with mean μ and variance σ^2. We want to test

$$H_0 : \mu = \mu_0 \qquad H_A : \mu \neq \mu_0$$

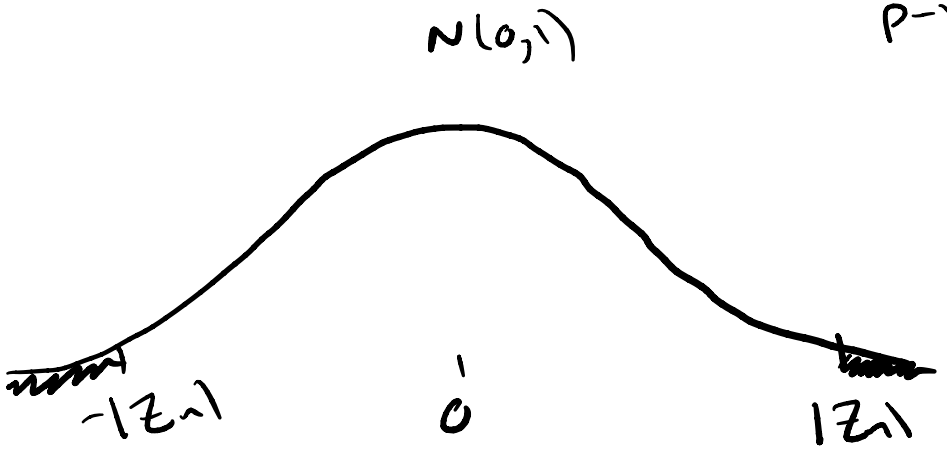
Test statistic: $Z_n = \dfrac{\overline{X}_n - \mu_0}{\sigma/\sqrt{n}}$

Rejecting: Reject H_0 when $|Z_n| > z_{\alpha/2}$

p-value: How do we calculate a p-value?

p-value $= 2\Phi(-|Z_n|)$

$N(0,1)$ cdf



Wald test for one parameter

Let $\theta \in \mathbb{R}$ be a parameter of interest, and let $\hat{\theta}_n$ be an estimator s.t. $\frac{\hat{\theta}_n - \theta}{s_n} \xrightarrow{d} N(0,1)$

for some sequence s_n $(s_n^2 \approx var(\hat{\theta}_n))$

To test $H_0: \theta = \theta_0$ vs. $H_A: \theta \neq \theta_0$ at level α

- let $Z_n = \frac{\hat{\theta}_n - \theta_0}{s_n}$

- reject when $|Z_n| > z_{\frac{\alpha}{2}}$ ← upper $\frac{\alpha}{2}$ quantile of $N(0,1)$

- p-value $= 2\Phi(-|Z_n|)$

↑ $N(0,1)$ cdf

Wald test for one parameter: examples

- Population mean: $H_0 : \mu = \mu_0$

$$\hat{\theta}_n = \bar{X}_n \quad , \quad \theta_0 = \mu_0 \quad , \quad S_n = \frac{\sigma}{\sqrt{n}} \text{ or } \frac{s}{\sqrt{n}}$$

$$s = \sqrt{\frac{1}{n-1} \sum_i (x_i - \bar{x})^2}$$

- Population proportion: $H_0 : p = p_0$

$$\hat{\theta}_n = \hat{p}_n \text{ (sample proportion)} \quad , \quad \theta_0 = p_0$$

$$S_n = \sqrt{\frac{\hat{p}_n(1-\hat{p}_n)}{n}} \quad \text{or} \quad \sqrt{\frac{p_0(1-p_0)}{n}}$$

- Regression coefficient: $H_0 : \beta_j = 0$

$$\hat{\theta}_n = \hat{\beta}_j \quad , \quad \theta_0 = 0 \quad , \quad S_n = \sqrt{\left[\underset{\sim}{I}^{-1}(\hat{\beta})\right]_{jj}}$$

jth diagonal entry of inverse Fisher info.

Testing multiple parameters

Logistic regression model for the dengue data:

$$Y_i \sim Bernoulli(p_i)$$

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 WBC_i + \beta_2 PLT_i$$

Researchers want to know if there is any relationship between white blood cell count or platelet count, and the probability a patient has dengue.

Question: What hypotheses should they test?

$H_0: \beta_1 = \beta_2 = 0$

$H_A:$ at least one of $\beta_1, \beta_2 \neq 0$

Testing multiple parameters

```
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    2.642      0.121  21.772        0
## WBC           -0.289      0.013 -21.533        0
## PLT           -0.007      0.001 -11.061        0
```

$H_0 : \beta_1 = \beta_2 = 0$

Can the researchers test their hypotheses using this output?

No: Each hypothesis tests $H_0: \beta_j = 0$

Wald tests for multiple parameters

$\mathcal{I}_1(\beta) = E[p_i(1-p_i)x_i x_i^T]$

$\mathcal{I}_n(\beta) = n\, E[p_i(1-p_i)x_i x_i^T]$

$\hat{\mathcal{I}}(\beta) = X^T W X$

$W = \mathrm{diag}(\hat{p}_i(1-\hat{p}_i))$

For the dengue example:

$$\widehat{\beta} = \begin{pmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \\ \widehat{\beta}_2 \end{pmatrix} \approx N(\beta, \mathcal{I}_n^{-1}(\beta)) \approx N(\beta, (X^T W X)^{-1})$$

```
m1 <- glm(Dengue ~ WBC + PLT, data = dengue, family = binom
coef(m1)
```

```
##  (Intercept)          WBC          PLT
##  2.641506279 -0.289290446 -0.006561464
```

```
vcov(m1)
```

$(X^T W X)^{-1}$

```
##                (Intercept)           WBC           PLT
## (Intercept)  1.471934e-02 -4.937020e-04 -5.125888e-05
## WBC         -4.937020e-04  1.804972e-04 -3.221337e-06
## PLT         -5.125888e-05 -3.221337e-06  3.518938e-07
```

Wald tests for multiple parameters

For the dengue example: $\widehat{\boldsymbol{\beta}} = \begin{pmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \\ \widehat{\beta}_2 \end{pmatrix} \approx N(\boldsymbol{\beta}, \underset{\sim}{I}_n^{-1}(\boldsymbol{\beta}))$

We want to test: $\begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $\qquad H_0: C\boldsymbol{\beta} = 0$

Rewrite: $\begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = C\boldsymbol{\beta}$

Here $C = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$$\begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = C\hat{\boldsymbol{\beta}} \approx N(C\boldsymbol{\beta}, C\,\underset{\sim}{I}_n^{-1}(\boldsymbol{\beta})C^T)$$

In general: hypotheses of the form

$$H_0: C\boldsymbol{\beta} = \gamma_0$$

$\uparrow$ vector under H_0

Wald tests for multiple parameters

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \gamma_0$$

For the dengue example:

```
C <- matrix(c(0, 1, 0,
              0, 0, 1), nrow=2, byrow=T)

C
```

```
##      [,1] [,2] [,3]
## [1,]    0    1    0
## [2,]    0    0    1
```

```
C %*% coef(m1)
```

```
##              [,1]
## [1,] -0.289290446
## [2,] -0.006561464
```

Wald tests for multiple parameters

For the dengue example:

```
C
```

```
##      [,1] [,2] [,3]
## [1,]    0    1    0
## [2,]    0    0    1
```

```
vcov(m1)
```

$(X^T W X)^{-1}$

```
##                (Intercept)           WBC           PLT
## (Intercept)  1.471934e-02 -4.937020e-04 -5.125888e-05
## WBC         -4.937020e-04  1.804972e-04 -3.221337e-06
## PLT         -5.125888e-05 -3.221337e-06  3.518938e-07
```

```
C %*% vcov(m1) %*% t(C)
```

```
##               [,1]          [,2]
## [1,]  1.804972e-04 -3.221337e-06
## [2,] -3.221337e-06  3.518938e-07
```

Test statistic and p-value

- $H_0 : \mathbf{C}\beta = \gamma_0$
- $\mathbf{C}\widehat{\beta} \approx N(\mathbf{C}\beta,\ \mathbf{C}\mathcal{I}^{-1}(\beta)\mathbf{C}^T)$
- Want to turn $\mathbf{C}\widehat{\beta}$ into a scalar test statistic

From HW: if $x \sim N(\mu, \Sigma)$, $x \in \mathbb{R}^q$ then $\Sigma^{-\frac{1}{2}}(x-\mu) \sim N(0, I)$

$$(x-\mu)^T \Sigma^{-1} (x-\mu) \sim \chi^2_q$$

<u>Under H_0</u>: $C\hat{\beta} \approx N(\gamma_0,\ C\mathcal{I}_n^{-1}(\beta)C^T)$

$$\Rightarrow \quad (C\hat{\beta} - \gamma_0)^T (C\mathcal{I}_n^{-1}(\beta)C^T)^{-1} (C\hat{\beta} - \gamma_0) \approx \chi^2_q$$

$q = \dim(\gamma_0)$

e.g. $H_0 : \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $\quad q = 2$

Example

$H_0 : \mathbf{C}\beta = \gamma_0$

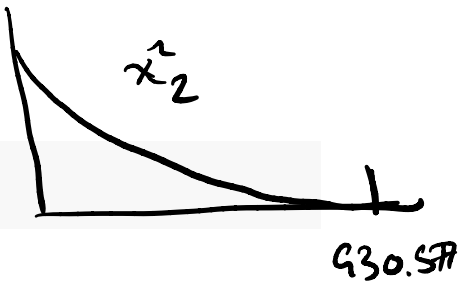
$$(\mathbf{C}\widehat{\beta} - \gamma_0)^T (\mathbf{C}\mathcal{I}^{-1}(\beta)\mathbf{C}^T)^{-1} (\mathbf{C}\widehat{\beta} - \gamma_0) \approx \chi^2_q \qquad \text{under } H_0$$

```
C %*% coef(m1)
```

```
## [1] -0.289290446 -0.006561464
```

```
test_stat <- t(C %*% coef(m1)) %*%
  solve(C %*% vcov(m1) %*% t(C)) %*%
  (C %*% coef(m1))
test_stat
```

```
##          [,1]
## [1,] 930.5777
```




```
pchisq(test_stat, 2, lower.tail=F)
```

```
##                [,1]
## [1,] 8.464858e-203
```

← p-value ≈ 0